Hieu Nguyen

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Summary

Engineer building and operating production data infrastructure. As CTO at Plenarius, operates a PostgreSQL platform over roughly 200,000 papers, tuning storage, indexing and query plans. Reduced an author profile from 7.07 s to sub-second through a partial covering index. At Sleep Sage, develops tested document-processing workflows. Prior production work at Viettel Post and FPT Software. Completed a Master of Artificial Intelligence at QUT in November 2026.

Skills

- **Data platforms:** PostgreSQL, pgvector, halfvec and binary quantisation, HNSW indexing, query plan tuning, schema migrations, MongoDB, job queues.
- **Pipelines and modelling:** Python, Go, SQL, batch and streaming ingestion, large-scale text processing, data quality checks, embedding models, predictive modelling, anomaly detection, RAG.
- **Cloud & infrastructure:** Azure (Functions, Cosmos DB), AWS (EC2, S3), Hetzner, Cloudflare (WAF, R2 object storage), Caddy, systemd.

Experience

Chief Technology Officer Sep 2026 – Present
Sleep Sage

- Lead technical development of an education app that turns PDFs and text into source-grounded concepts, user-approved learning cues and private speech clips.
- Improved React Native and Expo session editing, audio playback and first-night setup, and strengthened FastAPI processing with regression tests for validation, authentication and playback.

Chief Technology Officer Apr 2026 – Present
Plenarius (formerly Calimac)

- Owned the core PostgreSQL database tier, actively tuning query execution behaviour and managing schema migrations directly without relying on framework defaults.
- Analysed query execution plans and redesigned access paths, uncovering and fixing a pagination correctness bug caused by duplicate ordering timestamps while systematically accelerating high-traffic database endpoints.
- Evaluated multi-stage vector search designs combining dimensional reduction, quantised storage and exact re-scoring, comparing index footprint with retrieval quality.

AI Researcher Jan 2026 – Jun 2026
Cognilaw

- Evaluated and improved a hallucination-resistant RAG pipeline for legal workflows, testing retrieval strategies, grounded generation and answer validation using retrieval and response-quality metrics.

Research Assistant Sep 2025 – Nov 2026
Queensland University of Technology

- Evaluated and engineered an agentic person re-identification system combining appearance, body composition, gait and physical attributes through confidence-weighted voting.
- Cleaned and prepared biomedical literature datasets, then fine-tuned transformer models for protein-protein interaction relation extraction.

AI Software Engineer Apr 2023 – Nov 2024
Viettel Post

- Integrated Whisper fine-tuned on Vietnamese with BERT classifiers for offensive-language detection and mood prediction, achieving 90% accuracy in production.
- Trained an Extra Trees classifier for customer churn prediction, reaching 80% accuracy.

AI Software Engineer Oct 2021 – Mar 2023
FPT Software

- Contributed to FPT's US-patented Sound AI acoustic anomaly-detection platform, deployed on a Samsung factory floor. Built its Python backend on Azure Functions, supporting model serving, CI/CD and production monitoring.

Research Intern, Computer Vision Apr 2021 – Jul 2021
Asilla

- Fine-tuned and deployed real-time human pose estimation on Jetson Nano with TensorRT, using INT8 quantisation and structured pruning to improve efficiency by 20% under edge-compute constraints.

Projects & Awards

Planner and executor collaboration Manuscript in preparation
Stack: Python, AppWorld, BFCL, Agent evaluation

Co-author with Kien Nguyen Thanh at QUT of *How Should a Stronger Planner Help a Smaller Agent? Preregistered Tests of When to Hand Over and What to Send.* Built tool-using agent orchestration and paired replay evaluations comparing handover, advice and direct-action strategies with statistical analysis and cost-quality trade-offs.

Local AI Workflow Platform Presented | AgentCamp 2026, Brisbane Edition
Stack: Python, LangGraph, FastAPI, React Flow, Pydantic, YAML

Local-first platform for authoring, running and visualising AI workflows from declarative YAML. Each workflow is validated through a Pydantic GraphSpec contract before being compiled into runnable graphs, with approval interrupts, rollback restore and eval coverage.

Capture the Narrative 1st Place | UNSW National AI Competition
Stack: LLMs, multi-agent simulation, AWS EC2, S3

Built generative-AI agents inside a simulated election environment as an adversarial exercise in how synthetic content shifts sentiment.

SafeMind Most Viable | TANDA GenAI Hackathon
Stack: Python, Flask, OpenAI API, MongoDB, python-docx

Voice-driven workplace incident capture that pre-fills WorkCover claim forms, compressing hours of paperwork for injured workers.

Education

Master of Artificial Intelligence 2025 – 2026
Queensland University of Technology | GPA 6.538/7.0 (current)

Completed November 2026.

Bachelor of Information Technology 2018 – 2021
University of Science and Technology of Hanoi

Volunteer

Vice President Oct 2025 – Oct 2026
QUT The Emerging Coders Hub (TECH)
Ran coding and IT workshops teaching beginners to build chatbots, personal websites and AI agents, and organised career events including mock interview sessions, progressing from General Executive (Jun–Oct 2025).

Project Manager Sep 2025 – Nov 2026
QUT AI & ML Society
Led 30+ members building a predictive analytics dashboard on real insurance data for claim-delay prediction and anomaly detection, stepping up from Project Officer (Aug–Sep 2025).